

SHEIN E-Commerce Database Design and Management

Tracie Huynh

Project Advisor: Professor Aravinda Garimella

Department: Data Analytics Program

University of Illinois at Urbana-Champaign

Section 1: Context and Business Background Information

Shein is a Chinese global fast fashion brand that prides itself on its cheap prices and large quantity of designs. Shein was founded in 2008 in Nanjing China by Chris Xu. It became the world's largest fashion retailer as of 2022. However, as consumers grew to be more ethically conscious, Shein began to come under fire for their poor worker conditions and lack of quality control. As of current, Shein has continued to dominate the market despite the pushback, however the future of Shein and whether the damage to their reputation will catch up to them, remains uncertain.

The team had a profound interest in fashion and app development and decided to create a database system for Shein app that included but was not limited to entities such as users, orders, favorites, etc.

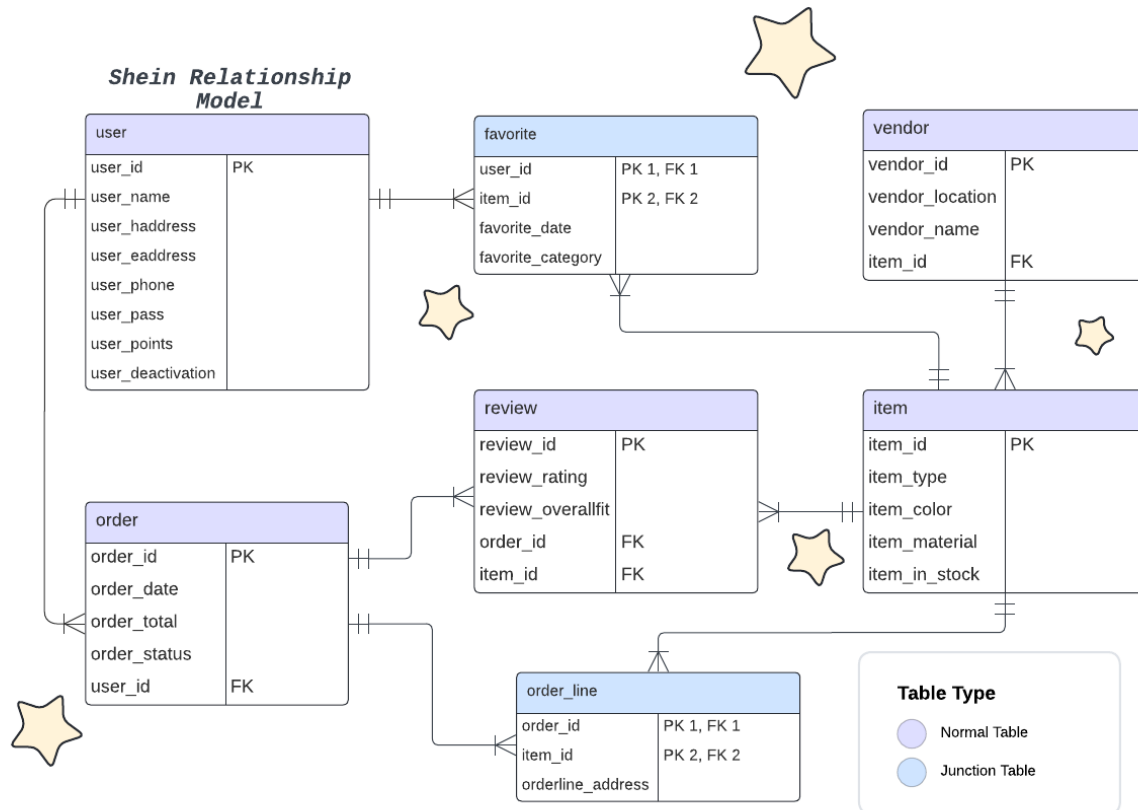
Section 2: Business Rules

We have a total of 8 business rules which are listed below:

- A user can buy multiple items, and an item can be bought by multiple customers.
- A user has one account, and an account belongs to one user.
- An order is associated with one user who placed it and a user can place multiple orders.
- An item can be marked as a favorite by multiple users and a user can put multiple items on their favorites list.
- An item can have only one producer and producers can have multiple items.
- An order can have multiple items and products can have multiple orders.
- Each item can have multiple reviews, but each review can only have one item.
- Items are shipped from multiple destinations and destinations can have multiple items.

Section 3: Relational Model

We had 7 tables in total, including the necessary junction tables: user, order, review, item, vendor, favorite, and order_line.



Section 4: Database Tables

The database helps SHEIN manage its products, customers, vendors, and orders. We have created several tables in our database. Below are the screenshots of our tables.

USER

user_id	user_name	user_haddress	user_eaddress	user_phone	user_pass	user_points	user_deactivation
00001	Montse Valencia	904 S 3rd St	monval@icloud.com	345-657-1046	Myname123	204	Active
00002	Sophia Korman	701 S Green Ave	skor@gmail.com	773-456-8046	Password	1000	Active
00003	Aimee Wang	999 W 8th St	happyman@gmail.com	222-222-2222	dudett22	128	Active
00004	Bob Walter	360 E Hollywood St	bobbywalter@hotmail.com	847-657-9019	Onetwothree	10	Active
00005	John Doe	55 N 5th St	sadman@gmail.com	333-333-3333	seventy2	99	Inactive
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

ORDER

order_id	order_date	order_total	order_status	user_id
00001	2022-11-13 22:22:22	2918.2	Delivered	00001
00002	2021-12-13 22:12:22	28.2	Delivered	00002
00003	2021-11-13 22:12:22	2.9	Not Delivered	00003
00004	2001-11-07 12:12:01	2.9	Delivered	00004
00005	2011-09-07 07:07:07	6	Delivered	00005
NULL	NULL	NULL	NULL	NULL

ORDER LINE

order_id	item_id	orderline_address
00001	45630052	904 S 3rd St
00002	27303243	701 S Green Ave
00003	23456389	904 S 3rd St
00004	34526792	360 E Hollywood St
00005	14790569	55 N 5th St
NULL	NULL	NULL

ITEM

item_id	item_type	item_color	item_material	item_in_stock
13148034	jean shorts	medium wash	denim	in stock
14790569	summer dress	baby blue	linen	in stock
23456389	mini skirt	black	rayon	out of stock
27303243	womens tops	black	viscose	in stock
34526792	baby tee	white	cotton	in stock
45630052	crewneck	white	cotton	in stock
NULL	NULL	NULL	NULL	NULL

FAVORITE

user_id	item_id	favorite_date	favorite_category
00001	23456389	2022-05-07 00:00:00	Bottom
00002	13148034	2020-06-13 02:02:02	Bottom
00002	23456389	2020-08-14 12:34:45	Bottom
00002	45630052	2020-07-08 14:56:09	Top
00004	45630052	2021-08-21 13:04:25	Top
00005	14790569	2023-09-30 08:45:56	Dress
NULL	NULL	NULL	NULL

REVIEW

review_id	review_rating	review_overallfit	order_id	item_id
122	4	Fit wells	00002	27303243
123	4	Perfect	00004	34526792
124	5	Loose	00001	45630052
125	3	Small	00005	14790569
126	2	Tight	00003	23456389
NULL	NULL	NULL	NULL	NULL

VENDOR

vendor_id	vendor_location	vendor_name	item_id
101	Italy	Happy Tees	45630052
102	Canada	Cold Sweaters Inc.	34526792
103	Japan	We are Crafty	23456389
104	Mexico	HeartJeans	13148034
105	Brazil	Sunny Smiles	13148034
106	Ukraine	BeSelf	27303243
NULL	NULL	NULL	NULL

Section 5: Code for Creation of Tables and Insertion of Data

CREATION:

Database –

```
create database mv30_sheinproject;
```

user table –

```
create table user(  
    user_id varchar(20) not null unique,  
    user_name varchar(35) not null,  
    user_haddress varchar(35) not null,  
    user_eaddres varchar(35) not null,  
    user_phone varchar(20) not null,  
    user_pass varchar(20) not null,  
    user_points float(20) not null,  
    user_deactivation varchar(20) not null,  
    primary key (user_id)  
);
```

order table –

```
create table vendor(  
    order_id varchar(20) not null unique,  
    order_date datetime,  
    order_total float(20),  
    order_status varchar(15),  
    user_id varchar(20),  
    primary key(order_id),  
    foreign key(user_id) references user(user_id)
```

);

item table –

```
create table item(  
    item_id varchar(15) NOT NULL UNIQUE,  
    item_type varchar(15) NOT NULL,  
    item_color varchar(15) NOT NULL,  
    item_material varchar(15) NOT NULL,  
    item_in_stock varchar(5) NOT NULL,  
    PRIMARY KEY (item_id)
```

);

vendor table –

```
create table vendor(  
    vendor_id varchar(20) not null unique,  
    vendor_location varchar(35) not null,  
    vendor_name varchar(35) not null,  
    item_id varchar(15) not null,  
    primary key(vendor_id),  
    foreign key(item_id) references item(item_id)
```

);

review table –

```
create table review(  
    review_id varchar(15) NOT NULL UNIQUE,  
    review_rating int NOT NULL,  
    review_overallfit varchar(10) NOT NULL,  
    order_id varchar(20) NOT NULL,  
    item_id varchar(15) NOT NULL UNIQUE,  
    primary key (review_id),
```

```
foreign key (order_id) references order_(order_id),  
foreign key (item_id) references item(item_id)  
);
```

order_line table –

```
create table order_line(  
    order_id varchar(20) not null,  
    item_id varchar(15) not null,  
    orderline_address varchar(35) not null,  
    primary key(order_id, item_id),  
    foreign key (order_id) references order_(order_id),  
    foreign key (item_id) references item(item_id)  
);
```

favorite table –

```
create table favorite(  
    user_id varchar(20) not null,  
    item_id varchar(15) not null,  
    favorite_date datetime,  
    favorite_category varchar(20),  
    primary key(user_id, item_id),  
    foreign key (user_id) references user(user_id),  
    foreign key (item_id) references item(item_id)  
);
```

DATA INSERTION:

User –

```
insert into user values('00001','Montse Valencia', '904 S 3rd St','monval@icloud.com', '345-657-1046', 'Mynamel23', 204, 'Active');
```

```
insert into user values('00002','Sophia Korman', '701 S Green Ave','skor@gmail.com', '773-456-8046', 'Password', 1000, 'Active');
```

```
insert into user values('00003', 'Aimee Wang', '999 W 8th St', happyman@gmail.com, 847-657-9019', 'Onetwothree', 10, 'Active');
```

```
insert into user values('00004','Bob Walter', '360 E Hollywood St','bobbywalter@hotmail.com', '847-657-9019', 'Onetwothree', 10, 'Active');
```

```
insert into user values('00004', 'John Doe', '55 N 5th St', 'sadman@gmail.com', '333-333-3333', 'seventy2', 99, 'Inactive');
```

order –

```
insert into user values('00001', '2022-11-13 22:22:22', 2918.2, 'Delivered', '00001');
```

```
insert into user values('00002', '2021-12-13 22:12:22', 28.2, 'Delivered', '00002');
```

```
insert into user values('00003', '2021-11-13 22:12:22', 2.9, 'Not Delivered', '00003');
```

```
insert into user values('00004', '2001-11-07 12:12:01', 2.9, 'Delivered', '00004');
```

```
insert into user values('00005', '2011-09-07 07:07:07', 6, 'Delivered', '00005');
```

item –

```
insert into item values(14790569, 'summer dress', 'baby blue', 'linen', 'in stock');
```

```
insert into item values(23456389, 'mini skirt', 'black', 'rayon', 'out of stock');
```

```
insert into item values(45630052, 'baby tee', 'white', 'cotton', 'in stock');
```

```
insert into item values(27303243, 'womens tops', 'black', 'viscose', 'in stock');
```

```
insert into item values(13148034, 'jean shorts', 'medium wash', 'denim', 'in stock');
```

vendor –

```
insert into vendor values('101','Italy', 'Happy Tees', 45630052);
```

```
insert into vendor values('102', 'Canada', 'Cold Sweaters Inc.', 34526792);
```

```
insert into vendor values('103', 'Japan', 'We are Crafty', 23456389);
```

```
insert into vendor values('104', 'Mexico', 'HeartJeans', 13148034);
```

```
insert into vendor values('105', 'Brazil', 'Sunny Smiles', 13148034);
```

```
insert into vendor values('106','Ukraine', 'BeSelf', 27303243);
```

review –

```
Insert into review value('122',4, 'Fits well', '00002', '27303243');
insert into review value ('123', 4, 'Perfect', '00004', '34526792');
insert into review value('124', 5, 'Loose' , '00001', '45630052');
insert into review value('125', 3, 'Small' , '00005', '14790569');
insert into review value('126', 2, 'Tight' , '00003', '13148034');
```

order_line –

```
insert into order_line value('00003', '13148034', '904 S 3rd St');
insert into order_line value('00004', '34526792', '360 E Hollywood St');
insert into order_line values('00001', '45630052', '904 S 3rd St');
insert into order_line values('00002', '27303243', '701 S Green Ave');
insert into order_line values('00005', '14790569', '55 N 5th St');
```

favorite –

```
insert into favorite values( '00001', 23456389, '2022-05-07', 'Bottom');
insert into favorite values('00002', 45630052, '2020-07-08 14:56:09', 'Top');
insert into favorite values('00002', 23456389, '2020-08-14 12:34:45', 'Bottom');
insert into favorite values('00002', 13148034, '2020-06-13 02:02:02', 'Bottom');
insert into favorite value('00004', 45630052, '2021-08-21 13:04:25', 'Top');
insert into favorite value('00005' , 14790569, '2023-09-30 08:45:56', 'Dress');
```

Section 6: Relevant Queries

1. List the order whose order id is 00001.

```
select * from order_ where order_id = '00001';
```

2. Count the number of items for each material group.

```
select item_material, count(*) from item group by item_material;
```

3. List the names of vendors whose name starts with 'H'.

```
select * from vendor where vendor_name like 'H%';
```

4. List orders that were made after 2020.

```
select order_id, order_date, user_id from order_ where order_date > '2020-01-01';
```

5. List all active users on Shein.

```
select user_id, user_name, user_eaddress, user_phone, user_deactivation from user where  
user_deactivation = 'Active';
```

Section 7: Limitations and Scope for Extension

Due to the project's limited time and resources, the team limited the database to the functions of the app system of Shein. Additionally, some relationships between entities may have been simplified due to the time constraints of the project.

Other ways to extend the project included having entities such as individual warehouses to track the supply chain of the delivery. This includes extending the database to track its sourcing agents such as showing each of the individual warehouses and the production process. This would be a great addition since that would allow Shein to keep track of each individual item's whereabouts in the production-line. Also, another aspect that could have been added is mapping out the revenue streams of the production, shipping, and sale of items to keep track of how much money Shein had.

